



Bgcse Agriculture - Notes

BGCSE Agriculture (University of Botswana)



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CRACKiT Tuition Botswana Term 3 National Examinations

Botswana General Certificate of Secondary Education

STUDENT
NAMES:

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CLASS:

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CRACKiT CENTE

**Agriculture
Paper 2
November 2023**

0583/02

2 hours

Students to answer on this question paper.

Additional materials:

1. Electronic calculator
2. Lined blank papers

READ THESE INSTRUCTIONS FIRST

Write your full names on this paper above.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams
or graphs.

Answer **all** questions in section A and answer four from section B

At the end of the exam, fasten all your work securely together.
Read and understand the questions before attempting them.

The total number of marks for this paper is 100.

FOR EXAMINER'S USE

Do not leave any

QUESTIONS	MARKS
Total Marks	
Percent age	%
Grade	

DO NOT TURN THE PAGE UNTIL YOU ARE TOLD TO DO SO.

Section A

Answer **all** the questions in the spaces provided.

1 Monoculture is an example of a farming practice.

(a) Describe what is meant by the term *monoculture*.

.....

.....

.....

..... [2]

(b) (i) State **two** benefits of monoculture.

1

.....

2

..... [2]

(ii) State **two** problems caused by monoculture.

1

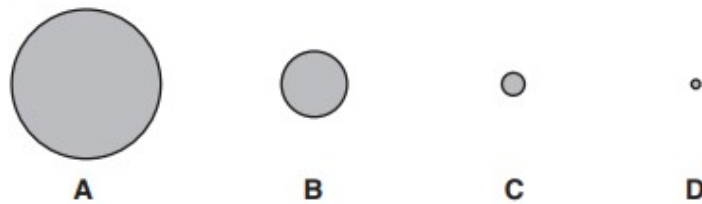
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2

..... [2]

[Total: 6]

- 2 (a) The diagram represents the order by size of clay, coarse sand, fine sand and silt particles.



Which letter represents a silt particle?

Answer **A, B, C or D** [1]

- (b) The table compares the properties of four soils **A, B, C** and **D**.

soil	A	B	C	D
water-holding capacity	medium	high	very high	very low
percentage of organic matter	4.5	8.0	7.5	2.0
rate of drainage	high	medium	low	very high
percentage of air	30	25	15	40

Only one of the soils is a sandy soil.

Which soil is the sandy soil?

Answer **A, B, C or D** [1]

- (c) (i) Describe how excess heat could affect the growth of seedlings in sandy soils.

.....

.....

.....

..... [2]

- (ii) Explain **one** way seedlings could be protected from the effects of excess heat.

.....

.....

.....

..... [2]

- (d) Suggest how frost damages crops.

.....

..... [1]

[Total: 7]

- 3 (a) Complete the paragraph using some of the following terms:

atmosphere

root hairs

stomata

vascular tissue.

Water is absorbed from the soil through the It then moves up through the plant in the before leaving the plant through the

[3]

- (b) Explain the effects of each of the following on the rate of transpiration:

humidity

.....

.....

.....

wind speed.

.....

.....

.....

[4]

- 4 (a) The photograph shows a forested area. The land must be prepared for the cultivation of a cash crop.



Describe how to clear and prepare this land to plant a cash crop.

.....

.....

.....

.....

(b) Name a crop.

.....

(i) Describe **two** signs that this crop is ready to be harvested.

1

.....

2

.....

[2]

(ii) Explain why a harvested crop should be kept in cool and dry conditions.

cool

.....

dry

.....

[2]

(c) Bread is an example of a product made from wheat.

State an example of a different product made from a named crop.

crop

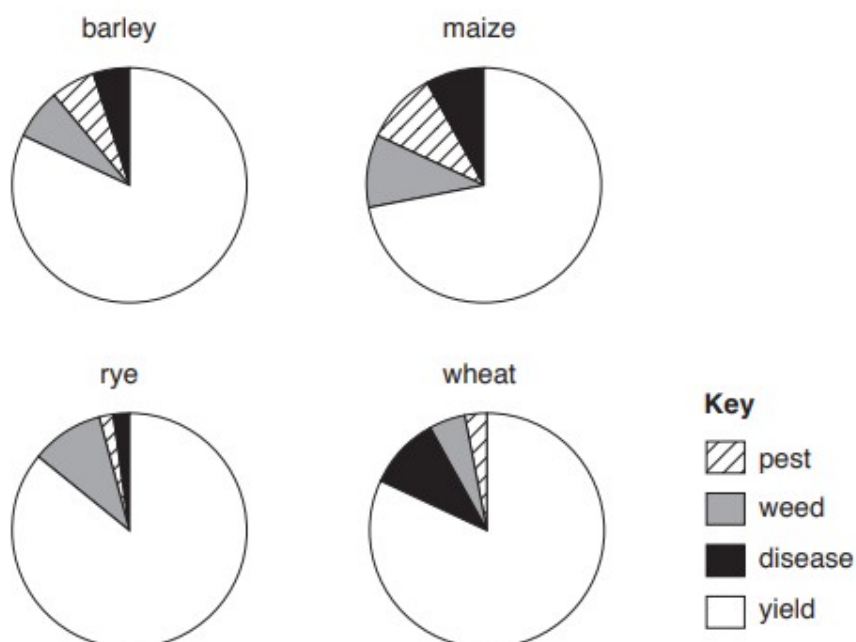
product

[1]

[Total: 8]

5 Pests have a major impact in reducing crop growth and yield.

- (a)** The pie charts show the estimated annual percentage yield and loss for four cereal crops worldwide. This loss is caused by pests, weeds and disease.



Use the pie charts to answer the following questions.

- (i)** Which crop suffered the largest percentage of pest damage?

- A** barley
- B** maize
- C** rye
- D** wheat

- (ii)** Which crop has the highest percentage total loss?

- A** barley
- B** maize
- C** rye
- D** wheat

Answer **A, B, C** or **D** [1]

(b) (i) Name **one** boring pest.

..... [1]

(ii) Explain why boring pests reduce crop yields.

.....
.....
.....
.....
.....
..... [3]

(c) Suggest why some farmers do **not** use chemical methods of pest control.

.....
.....
.....
..... [2]

[Total: 8]

- 6 (a)** Some crops are grown in a rotation.

Describe an example of a four-year crop rotation.

.....

.....

.....

..... [2]

- (b)** Explain, using an example, what is meant by the term *compound fertiliser*.

.....

.....

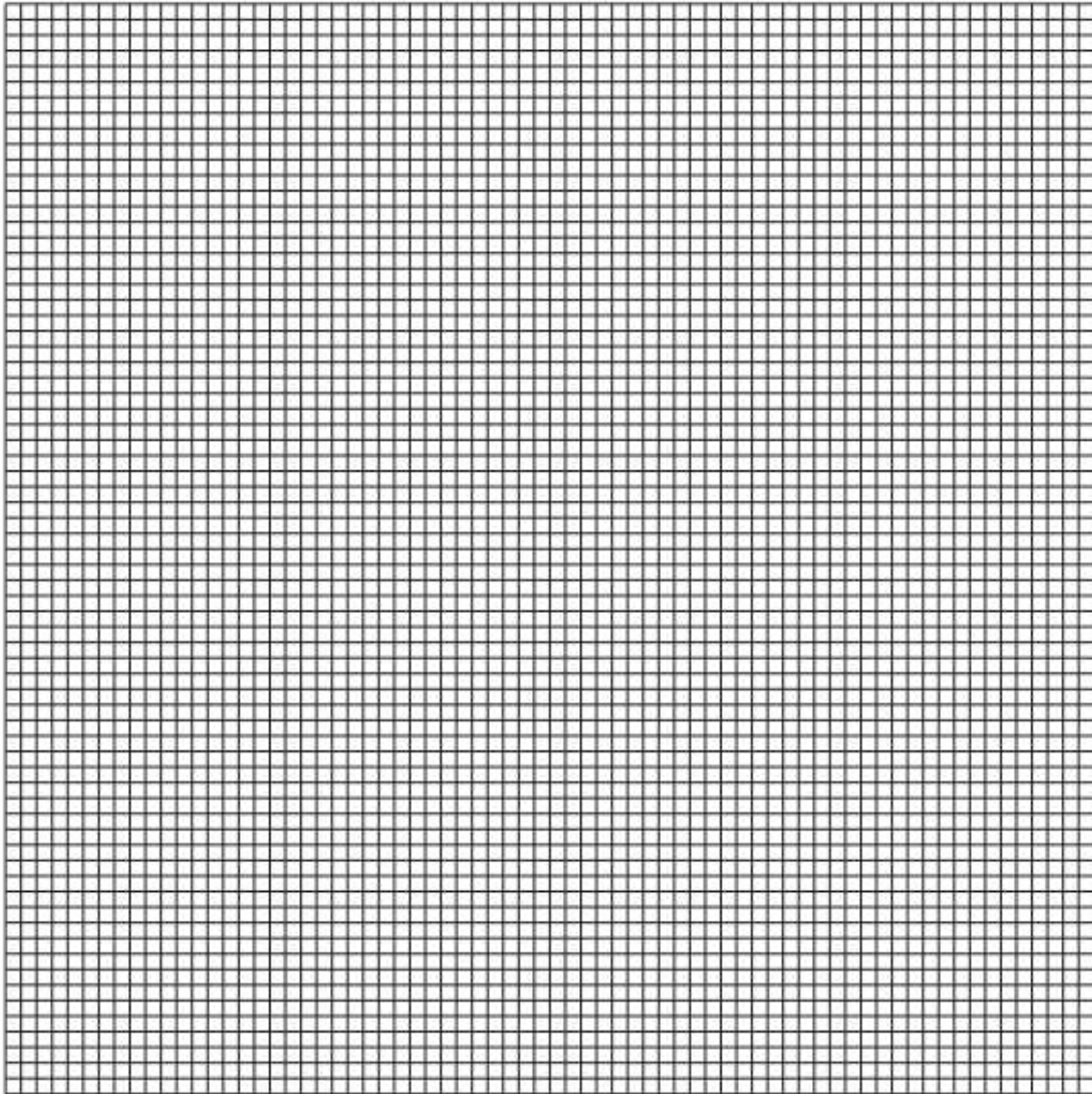
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..... [2]

- (c)** The table shows production records from a field in a farm that grew rice as a single, continuous crop.

year	rice yield per year /t
1	3.1
2	2.5
3	2.6
4	1.5
5	1.2

Draw a line graph to show the information from the table. Draw a line of best fit.



[3]

SECTION B

ANSWER THREE QUESTIONS ON THIS SECTION

Write your answers on the separate paper provided.

- (a) Describe how climate and topography affect the choice of farming practice. [4]
- (b) Describe how population growth can impact on land use in farming. [4]
- (c) Explain how to minimise soil erosion on sloping land. [7]

[Total: 15]

- (a) Describe what is meant by the term *photosynthesis*. [3]
- (b) Outline the process of gas exchange in leaves. [5]
- (c) Describe how plants transport and store food. [7]

[Total: 15]

- (a) Describe reasons why a soil may become acidic. [3]
- (b) Describe how the soil pH of a farm should be tested. [5]
- (c) Explain how crops absorb mineral salts from the soil. [7]

[Total: 15]

- (a) State what is meant by the term *sexual reproduction*. [3]
- (b) Name the parts of a bean plant flower and describe the function of each part. [7]
- (c) Explain the advantages of sexual reproduction compared to asexual reproduction in crops. [5]

[Total: 15]

- (a) Describe the reproductive system of a female mammalian farm animal. [4]
- (b) Describe the process of birth for a named mammalian farm animal. [5]
- (c) Explain the care needed for the mammalian farm animal in (b) from birth until weaning. [6]

[Total: 15]